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1. Overall Summary

This paper introduces DeLAMA, a novel decentralized and lifelong-adaptive multi-agent collaborative learning framework designed to overcome limitations in existing systems, such as reliance on centralized servers and static collaboration relationships, which hinder efficiency and robustness in dynamic environments. DeLAMA empowers agents to autonomously discover beneficial collaboration partners and adapt to evolving tasks. The approach integrates a decentralized graph structure learning algorithm for autonomous relationship inference and a memory unit to capture accumulated learning history with finite storage. By leveraging algorithm unrolling, DeLAMA combines the strengths of mathematical optimization and neural networks, allowing agents to 'learn to collaborate' and enhancing both expressive capabilities and computational efficiency. Theoretical analysis confirms communication efficiency and convergence properties. Experimental results on regression, image classification, and multi-robot mapping tasks demonstrate DeLAMA's superior performance, achieving a 98.80% reduction in MSE and a 188.87% improvement in classification accuracy compared to state-of-the-art baselines. The work provides a foundational technique for intelligent, decentralized, and dynamic multi-agent systems.

2. Main Idea

The central concept is DeLAMA, a decentralized and lifelong-adaptive multi-agent collaborative learning algorithm that enables agents to autonomously identify beneficial collaboration relationships and adapt to dynamically changing tasks over time, without relying on a central server. It achieves this by combining mathematical optimization and neural networks through algorithm unrolling.

3. Problem Being Solved

This paper addresses two major weaknesses in previous multi-agent collaborative learning systems: 1) the vulnerability and lack of robustness caused by reliance on a central server for coordination and collaboration, and 2) the inefficiency and inflexibility of predefined or static collaboration relationships in dynamic scenarios where tasks and agent relationships evolve over time. It also aims to improve collaboration efficiency by avoiding redundant and low-quality information from arbitrary collaborators.

4. Assumptions

- Training data arrives independently among agents and across time.
- Similar model parameters promote agents to collaborate.
- The adjacency matrix $W(t)$ is symmetric for simplifying Jacobi iteration.
- Standard supervised training loss (mean square error or cross-entropy) and linear model are used for analytical determination of the best Taylor expansion point.
- The function $h_b(\cdot)$ is an approximation function of $\text{ReLU}(\cdot)$.
- In the regression experiment, the collaboration relationship is easy to verify.
- Human collaboration strength can be quantified by rating scores (0 to 5), where a higher score means higher collaboration.
- Humans tend to collaborate with others who have similar model parameters.

5. Limitations

- The Taylor expansion in the local learning phase has potential approximation errors for non-linear functions, limiting the method's expressive capability for non-linear learning tasks.
- The collaborative learning system has many hyper-parameters, requiring significant effort for searching and tuning.
- The algorithm requires numerous iterations (single graph learning, message passing, and alternating convex search), leading to significant communication overhead and making it less practical for real applications.
- For harder collaboration problems with lower data quality, humans have difficulty distinguishing useful collaborators, suggesting that human collaboration mechanisms involve more complex logic than simple similarity, which the current model may not fully capture.

6. Results and Conclusion

DeLAMA significantly outperforms existing federated learning and decentralized optimization methods, achieving a 98.80% reduction in MSE and a 188.87% improvement in classification accuracy. It demonstrates superior performance in regression, image classification, and multi-robot mapping tasks. The approach effectively learns collaboration relationships, adapts to dynamic learning scenarios, and maintains performance even under low communication connectivity. Theoretical analysis confirms favorable convex optimization properties, quadratic convergence for collaboration inference, and linear convergence for model updating, ensuring communication efficiency. Human-involved experiments show DeLAMA's task performance is comparable to humans and its collaboration mechanism mirrors human cognitive collaboration, verifying its ability to discover suitable collaboration strategies and adapt to dynamic learning environments.

7. Real World Impact

This research has practical real-world applications in various domains, including vehicle-to-everything (V2X) communication-aided autonomous driving, multi-robot environment exploration, collaborative training of machine learning models across multiple clients, and smart healthcare. It serves as a foundational technique for developing intelligent, decentralized, and dynamic multi-agent systems.

8. Graphs and Figures Analysis

Figure 1: The illustration of our decentralized and lifelong-adaptive multi-agent collaborative learning system.

This figure visually represents the DeLAMA system. It shows six agents, each with model parameters, observations, and collaboration relationships. The system evolves over time (Timestamp 1, 2, 3), demonstrating how collaboration relationships dynamically adapt to time-evolving tasks. The legend clarifies the symbols: model parameters (blue circles), collaborating agents (red owls), different observations (green squares), and collaboration relationships (red dashed lines). The dynamic nature of tasks and relationships is central to the system's design.

Figure 2: Relationships among our lifelong collaborative learning and its related works: lifelong learning, federated learning, and multi-task relation learning.

This 2x2 matrix plot categorizes different learning paradigms based on two axes: "Dynamic scenario" (top row) vs. "Static scenario" (bottom row) and "Predefined collaboration relations" (left column) vs. "Learned collaboration relations" (right column). The top-left shows Lifelong learning (dynamic, predefined relations), bottom-left shows Federated learning (static, predefined relations), bottom-right shows Multi-task relation learning (static, learned relations), and the top-right shows Decentralized and lifelong-adaptive collaborative learning (DeLAMA) (dynamic, learned relations). This figure highlights that DeLAMA uniquely addresses both dynamic scenarios and learned collaboration relationships, distinguishing it from existing methods that typically handle only one or neither.

Figure 3: The decentralized and lifelong-adaptive multi-agent collaborative learning system.

This figure illustrates the three main iterative steps of the DeLAMA system: 1) Local learning, where each agent learns initial model parameters based on its own observations; 2) Collaboration relational inference, where agents transmit model parameters and learn collaboration relationships; and 3) Lifelong model update, where agents share parameters based on learned relationships and refine models. The figure shows agents with different observations, communication graph constraints, and collaboration relations. The iterative nature of the collaboration relational inference and lifelong model update steps is indicated by the "Iterate" arrow, visualizing the decentralized and adaptive learning process.

Figure 4: The unrolled network structure of the collaboration system with input data at one exact time.

This figure depicts the neural network architecture of DeLAMA, derived from unrolling the iterative optimization algorithm. It shows the flow from input data for each agent to the final output model parameters. Local Learning processes input data ($X_i(t)$, $Y_i(t)$) via a backbone network (Φ_β) to initialize model parameters. Collaborative Relational Inference then uses these parameters and the communication graph to infer collaboration relationships ($W(t)$) through a graph learning layer. Finally, Lifelong Model Update refines the model parameters over multiple layers using the learned collaboration relationships. The figure highlights learnable hyperparameters and the integration of mathematical optimization steps into a deep neural network structure.

Table 1: Relations to previous task settings, including lifelong learning, multi-task learning, and federated learning.

This table compares DeLAMA with existing learning paradigms (Lifelong learning, Multi-task learning, Federated learning) across three dimensions: "Dynamic scenario", "Decentralized mechanism", and "Number of agents". It shows that Lifelong learning is dynamic but centralized, Multi-task learning and Federated learning are static and centralized. DeLAMA (Ours) is unique in simultaneously supporting dynamic scenarios (✓), a decentralized mechanism (✓), and multiple agents (N), addressing the limitations of previous methods.

Table 2: Performance comparison for the regression problem.

This table compares the Mean Squared Error (MSE) of DeLAMA against various federated learning and decentralized optimization methods for a regression problem at timestamps $t=1$ and $t=10$. DeLAMA (2.3377 at $t=1$, 0.0719 at $t=10$) consistently achieves the lowest MSE, significantly outperforming all other methods. Ablation studies (DeLAMA-WC: without collaboration, DeLAMA-WM: without lifelong-adaptive learning) show that both collaboration and lifelong adaptation are crucial for performance, especially at later timestamps, demonstrating DeLAMA's superior performance and the importance of its core components in dynamic regression tasks.

Table 3: The evaluation of the collaboration performance compared with other centralized collaboration mechanisms.

This table evaluates DeLAMA's collaboration performance on regression tasks, comparing it with other centralized and decentralized structural learning methods under different communication graph constraints (FC, ER, BA graphs with varying connectivity levels). Metrics include Graph Mean Square Error (GMSE) and MSE at $t=1$, $t=5$, and $t=10$. DeLAMA consistently achieves the lowest GMSE and MSE across all communication structures and timestamps (e.g., FC graph: GMSE=0.0721, MSE=0.0719 at $t=10$), significantly outperforming baseline methods and performing close to the "Oracle" (ground truth) method. This confirms DeLAMA's effectiveness in learning accurate collaboration relationships and improving task performance, even under varying communication constraints.

Figure 5: Visualization of the learned graph structure at time $t = 3$ compared with the oracle collaboration structure.

This figure visualizes the learned collaboration graph structures at time $t=3$ for different methods (GL-LogDet, GLasso, MTRL, L2G-PDS, NS, DeLAMA) compared to the Oracle (ground truth) structure. Each grid represents the collaboration strength between agents. DeLAMA's learned graph structure closely resembles the Oracle, showing strong, smooth, and accurate edge weights that correctly identify collaboration patterns. Other baseline methods show less accurate or more noisy collaboration structures, indicating they struggle to identify beneficial relationships as effectively. This qualitative analysis supports DeLAMA's ability to discover suitable collaboration relations.

Figure 6: A collaboration result of two groups of agents that are performing two different regression tasks.

This figure shows two parts: (a) Functions learned by each agent, plotting the learned regression functions for six agents over time ($T=1$ to $T=5$). Each agent's learned function (colored lines) gradually approaches the true underlying function (black line), demonstrating lifelong learning and adaptation. (b) Time evolving collaboration structure, visualizing the collaboration graph among agents from $T=1$ to $T=5$. Initially, connections are broader, but as learning progresses, the structure refines into two distinct groups corresponding to different tasks, illustrating DeLAMA's dynamic evolution of collaboration relationships.

Table 4: The system performance analysis on classification tasks of four different settings.

This table presents the average classification accuracy (Acc) for MNIST and CIFAR-10 datasets at timestamps $t=1$, $t=5$, and $t=10$, comparing DeLAMA with lifelong learning, federated learning, and decentralized optimization methods. DeLAMA (e.g., MNIST Acc=99.51% at $t=10$, CIFAR-10 Acc=76.03% at $t=10$) significantly outperforms all other categories of methods. Ablation studies (DeLAMA-WC, DeLAMA-WM) show that both collaboration and lifelong adaptation are critical for high accuracy, especially at later timestamps, highlighting DeLAMA's superior ability to remember past training data and adapt to new tasks in classification scenarios.

Table 5: The system collaborative learning performance of different collaboration relationship learning methods.

This table compares DeLAMA's performance on classification tasks (MNIST and CIFAR-10) with various collaboration relationship learning methods, using GMSE and classification accuracy at $t=1$, $t=3$, and $t=5$. DeLAMA consistently achieves the lowest GMSE and highest accuracy across both datasets and timestamps (e.g., MNIST: GMSE=0.0855, Acc=98.03% at $t=5$; CIFAR-10: GMSE=0.1475, Acc=72.57% at $t=5$). It outperforms baseline methods and performs very close to the "Oracle" method, reinforcing DeLAMA's effectiveness in learning accurate collaboration graphs and improving classification performance.

Figure 7: Example of the evolving collaboration graphs.

This figure visualizes the time-evolving collaboration graphs for MNIST and CIFAR-10 datasets at different timestamps ($T=1$, $T=3$, $T=5$, $T=7$) compared to the Oracle (ground truth) structure. Initially ($T=1$), agents might collaborate more broadly. As time progresses ($T=3$, $T=5$, $T=7$), DeLAMA refines the collaboration relationships, cutting off less useful connections and strengthening beneficial ones, making the learned graphs increasingly similar to the Oracle structure. This demonstrates DeLAMA's dynamic adaptation of collaboration relationships based on evolving observations.

Figure 8: The simulation result of robot mapping at $t = 12$ based on the CubiCasa5k room Dataset.

This figure shows the 2D robot mapping results at $t=12$ for different decentralized collaborative model training approaches (Truth Map, DeLAMA, DeLAMA-WC, DSGD, DSGT, DiNNO). Black represents empty space, and white represents occupied space. DeLAMA's map is the most accurate and complete, closely matching the Truth Map. DeLAMA-WC (without collaboration) shows a less complete map, highlighting the benefit of collaboration. Other decentralized methods produce significantly less accurate or incomplete maps, qualitatively demonstrating DeLAMA's superior performance in multi-robot mapping tasks, enabling more complete and accurate environmental exploration through collaboration.

Table 6: The agents' average F1-score of the multi-robot mapping task.

This table compares the average F1-score of DeLAMA with other decentralized collaborative model training approaches for the multi-robot mapping task at timestamps $t=1$, $t=6$, and $t=12$. DeLAMA (e.g., F1=95.31% at $t=12$) achieves the highest F1-score, significantly outperforming DSGD, DSGT, and DiNNO. DeLAMA-WC (without collaboration) has a lower F1-score (74.92% at $t=12$), indicating that collaboration is crucial for performance in this task. This table quantitatively confirms DeLAMA's effectiveness in multi-robot mapping scenarios.

Table 7: The average performance of human collaboration compared with DeLAMA.

This table compares the average MSE of human collaboration with DeLAMA under different data quality levels (Low, Medium, High) at timestamps $t=1$, $t=2$, and $t=3$. DeLAMA consistently achieves lower MSE than human collaboration across all data quality levels and timestamps (e.g., High data quality: DeLAMA MSE=0.0112 at $t=3$ vs. Human MSE=0.0270 at $t=3$). Both human and DeLAMA performance improve with higher data quality and over time, suggesting that DeLAMA can achieve comparable or even superior task performance to human collaboration in cognitive learning scenarios.

Figure 9: The human evaluation of collaboration relationships with each other.

This figure shows box plots of the logarithm of model parameter distance (y-axis) against human rating scores (x-axis, 0 to 5, representing collaboration strength) for different data quality levels (High, Medium, Low). The plots show a negative correlation: as human rating scores (collaboration strength) increase, the logarithm of model parameter distance decreases, verifying the assumption that humans tend to collaborate with others who have similar model parameters. For lower data quality, the distribution of distances for higher human scores is wider, suggesting that distinguishing useful collaborators becomes harder, implying more complex collaboration logic beyond simple similarity.

Figure 10: A time-evolving learning task example of two agents sampled from the regression task dataset.

This figure shows two rows of plots. The top row displays the learned regression functions for two agents (Agent 1 and Agent 2) at different timestamps ($T=1$ to $T=5$), along with the ground truth function. The bottom row shows the corresponding scatter plots of the training data. As time progresses, the agents' learned functions (colored lines) gradually converge towards the ground truth function (black line), demonstrating that their knowledge of the target function improves with an increasing number of sampling intervals. The scatter plots illustrate the limited and evolving nature of the data available to each agent at different times.

Figure 11: The human-machine interaction framework used for the human-involved experiment.

This figure illustrates the hierarchical framework for the human-involved experiment. The User Interface Layer shows human users visually interacting with "virtual agents" (proxy machines). The Back End shows the core DeLAMA collaboration algorithm running, with "proxy machines" imitating human behaviors to enable interaction between humans and the DeLAMA system. The Interaction Phase depicts agents (human and virtual) observing each other's behaviors and submitting their understandings. This framework provides a platform for real human evaluation of DeLAMA's collaboration mechanisms.

Figure 12: The login page of the human-involved experiment, contains a user guide for the participants to learn how to play the game.

This figure shows a screenshot of the login page for the human-involved experiment. It includes fields for User ID and Username, along with a user guide explaining the game's objective (linear regression to minimize loss with partial, noisy data) and how agents can modify their answers by referring to others' results. It also shows small example plots of data at different times.

Figure 13: The local learning page of the human-involved experiment.

This figure shows a screenshot of the local learning page from the human-involved experiment. A participant sees a plot of scatter points (representing their local data) and draws a line to represent their guess of the regression line. The drawn line is then transformed into a linear equation with two parameters, which is displayed below the plot. This page captures the initial, individual learning step of each participant.

Figure 14: The aggregation page of the human-involved experiment, which contains other participants' results and the score rated by each agent.

This figure shows a screenshot of the aggregation page from the human-involved experiment. It displays the regression lines drawn by other participants and the scores (stars) rated by each agent. This page allows participants to view and consider others' learning results before modifying their own, simulating the collaborative aspect of the system. The evaluation loss for the participant's performance is also shown.

Figure 15: The final page of the human-involved experiment with a leaderboard, each participant can look back to historical learning results.

This figure shows a screenshot of the final page of the human-involved experiment. It includes a "Leaderboard" displaying average performance metrics and a "Result Log" showing the participant's historical learning results (evaluation loss over time) compared to the ground truth regression target function. This page provides a summary and allows participants to review their performance and the overall game outcome.